

12 Form, Cause, and Explanation in Biology

A Neo-Aristotelian Perspective

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1 Introduction

Aristotle's theoretical division of the concept of causation has been an unquestionably influential and important component in the analytic tradition of metaphysics. The most contentious aspect of that four-fold partition is arguably its inclusion of what Aristotelians termed formal causation—or causation 'by means of' the essence of an entity. Not only is this sort of causation the one most intimately tied up with the wider metaphysical assumptions of Aristotle's hylomorphic framework, but the distinctive role it purports to play in causal explanations has increasingly been seen as, at best, entirely superfluous and at worst, fundamentally suspect.

Formal causation, for Aristotle, occurs when the *essence* of an entity is responsible for bringing about that entity's possession of some feature, or property; it therefore features in explanatory schemas in which invoking an entity's membership to a particular *kind* provides an answer to specific 'why?' (*aitia*) questions involving its features, or properties.¹ This explanatory project works because, in the Aristotelian framework, essence is an ineliminably *teleological*, or 'goal-directed'. While essence might be rendered as a sort of inert and lifeless logical definition of 'what it is to be' that entity (or an entity *of that kind*), an entity's essence is an *intrinsic* and fundamentally *dynamical* feature—a 'principle of activity' (*archê tou energeia*)—which functions as the ontological ground of that entity's *de re* modal profile with respect to its morphology (its *morphê*).²

Of course, aside from there being such a thing as form or essence which matches this description, the viability of this project crucially depends upon there being a coherent sense in which teleological 'directedness' can be attributed any genuinely *causal* import. Many modern metaphysicians will likely readily affirm the causal impetus inherent in Aristotle's other, by now familiar notions of *material* and *efficient* causation, but the conceptual resources with which one might understand those notions aren't available in the case of formal causation. For according to Aristotle, the form of an entity is (1) a 'holistic' feature which is in some sense irreducible to or 'over and above' that entity's mereological

make-up,³ and is (2) incapable of both moving (other things) and being moved (by other things).⁴ Clearly, if there is an ontological ground of an entity's belonging to a particular natural kind which in some way causally influences the sort of properties that entity possesses, it must do so in a wholly unique (and utterly unfamiliar) fashion.

Whether an entity's features can be caused by—or else informatively explained by means of an appeal to—its possession of an intrinsic teleological impetus—namely, its form—was, for a long period in the history of philosophy, not viewed as an open question. The persisting academic influence of 'The Philosopher' aside, the conceptualisation of the world as 'carved at the joints' by natural kinds was widely accepted in the natural sciences, and most enthusiastically so in Aristotle's own favoured branch—biology. Organisms, of course, were most naturally amenable to Aristotelian analysis: if the activities of any entities could be explained teleologically, it was those that constituted the *living* world. With the advent of the scientific revolution in the 17th century, however, the Aristotelian categorisations of causation—and their associated metaphysical underpinnings—were rejected, and replaced. And while the revisionary wave of the new mechanistic metaphysic of Descartes, Gassendi, and Newton that had swept through every natural science was held back at the bay of biology for some time, the 19th century advent of evolutionary theory saw it too awash with anti-Peripatetic tides.

Today, the biological sciences—led by the substantially expanded theoretical prowess of evolutionary theory wrought by the 'Modern Synthesis' of population genetics and natural selection—have fully 'mechanized' the operations of organisms and so all but eliminated the Aristotelian concept of *natural kind* and with it, the utility of *formal causation*. However, although in contemporary evolutionary biology Aristotelianism is largely viewed as an ancient artefact of an essentially pre-empirical age, it is currently enjoying a small, but spirited philosophical revival as its perspective on the modality of morphology and the role of organisms in the process of selection are being appreciated anew in novel theoretical contexts.⁵ To my mind, this is no accident: this resurgence has been to a considerable degree inspired by a corresponding revolution in evolutionary theory—the product of the most modern theoretical synthesis, the conceptual framework of evolutionary-developmental biology, or, for short, 'evo-devo'. In this chapter, by examining the roots of the rejection of form implicit in some of the core tenets of evolutionary theory, I show how the explanatory principles of the new science of evo-devo might contain the seeds of its rebirth.

2 The Mechanisms of Evolution

The long reign of the Aristotelian framework in the history of both metaphysics and the natural sciences has been exceedingly well documented.

Its several successes and insights notwithstanding, its eventual descent into theoretical territories which were widely viewed as consisting of insubstantialities, trivialities, and uninformed speculations is also written in the record of history. The perceived problems with the Aristotelian framework were compounded by the fact that novel proposals for the fundamental restructuring of the natural sciences were few and far between and, due to its pervasiveness and culturally well-entrenched status, such proposals necessarily took place in the context of its core principles and thus typically constituted little more than a cautious and minor revision of them.

This conceptual stalemate was, however, effectively ended in the wake of what is now known as the 'scientific revolution' of the 16th and 17th centuries. Led by a small but influential network of philosophers-*cum*-physicists, new epistemological foundations for empirical knowledge and novel conceptual frameworks for classifying and comprehending large classes of natural phenomena were developed. Although there were many, the innovation with the most far-reaching theoretical implications was arguably the reconceptualisation of what causation consists in. As the causal framework underpinning the fundamentally Aristotelian science of the 'schoolmen' was increasingly understood as lacking sufficient explanatory prowess, its central theoretical posits became the subjects of intense philosophical scrutiny. This led, eventually, to a spirited rejection of an ontology of powers, or potentialities—for these were mere scholastic or occult qualities which could not be properly expressed in nor comprehended by the newly crowned 'language in which God has written the universe' (Mathematics, according to Galileo) whose true natures were empirically inaccessible, if not necessarily impenetrable.⁶ Likewise, teleology too was removed from its principled position in nature's operations—explaining the activities of an entity as the ends or effects of an intrinsic, goal-directed impetus became widely viewed as at best an essentially otiose and uninformative enterprise, and at worst, as entailing a mild form of panpsychism.⁷

Given that the Aristotelian concept of form is that of a *teleologically directed power*, it's no surprise that it too fell under the revolution's conceptual guillotine. Explaining the causes of natural actions, in the framework of the 'new science', was seen as requiring the conceptual tools of *quantitative*, rather than *qualitative*, analysis. In particular, explanations of a system's activities via qualitative description—invoking the qualities, powers, or forms that systems possessed—were regarded as epistemologically inferior to explanations which utilized the methodology of quantitative decomposition—ones that dissected systems into the sets of elements which constituted their mereological make-up and laid bare the mathematically and geometrically tractable relations among their members. Importantly, the theoretical successes of analyses which engaged in this empirical process of quantitative decomposition

seemed to indicate not only that the qualitative features of entities lacked sufficient explanatory power, but that whatever prowess they possessed was fundamentally misplaced: explaining the activities of entities in terms of the quantitative features of and relations among their parts required an appeal to the various forces of *extrinsic* compulsion *acting on* those parts (and the character of the laws that govern them) rather than to the natures of the intrinsic impetus *inherent in* them. In short, the utilization of the methodological principles which guide this form of explanation suggest the adoption of a broadly *mechanistic* perspective on the nature of causation.

Though the ascendance of this mechanical philosophy in the context of the scientific revolution is a complex and multifaceted affair (Craver and Darden 2005; Roux 2018) what I want to highlight here is simple: the metaphysical postulates of this mechanistic world are radically anti-Aristotelian. From the mechanistic perspective, explaining the activities of the entities which populate the natural world requires conceptualising them as the products primarily of those entities' places within the wider context in which they happen to be embedded and the relations they bear to one another therein. From a metaphysical point of view then, the entities of this world are essentially *passive* participants in the production of effects whose activities are fundamentally *extrinsically* determined by the combination of laws of nature and *contingent* circumstances (Ellis 2001, 2002; Groff 2013). Having been effectively emptied of their essential, intrinsic natures (of the sort to which Aristotelian metaphysics makes explanatory appeal), the entities of this world, 'however absolute and entire they seem in themselves', as Locke (*Essay* IV.6, sec. 11) put it, 'are [each individually] but retainers to other parts of nature.'

While the conceptualisation of the natural world was being thoroughly mechanized during the scientific revolution, the biological realm remained largely shielded from such revision and its study was widely regarded as a 'special science'.⁸ As Aristotelian substances *par excellence*, organisms were still viewed as having forms—teleologically rich causal structures which ontologically imbued their bearers with specific natural kind identities. The reason is clear: the empirical and theoretical successes of the programme of mechanization notwithstanding, the phenomenon of organismal development still seemed best explainable by appealing to intrinsic, active, and non-accidental forces. For instance, in a pre-genetic age, the 'miracle' of reproductive fidelity—the fact that an organism and its progeny were morphologically alike (and could not be otherwise)—seemed to be the result of the working of intrinsic, goal-directed forces. Likewise, the specialised suitability of organisms and their parts—the evident 'harmony' between morphology and function, as exhibited by organs being 'fit' to particular purposes and organisms to specific environments—cried out for a teleological explanation if anything did.⁹ What these central and well documented biological

phenomena (among many others) highlighted was that there simply were no plausible *mechanistic* models with which to conceptualise the goal-directed morphogenesis of organismal features, and no known laws of nature which might produce them; the absence of which eventually paved the way for the rise of various forms of vitalism in the following centuries (Normandin and Wolfe 2013; Chen 2018).¹⁰

With the advent of the theory of evolution in the mid-19th century, however, another revolution took place—this time in the biological sciences. Led by the pioneering work of Darwin and Wallace, the discovery of new *mechanisms* suggested new *causes*: the combination of phenotypic variation, natural selection, and ‘deep time’ (the wide-scope time-scale metric mapped by geologically based analyses of the planet’s sedimentary record) offered novel and powerful explanatory insight into the origin of the generative specificities of organismal morphology (Ruse 1979). The morphological profiles of organisms—their various and particular anatomical (‘interior’) and eidonomical (‘exterior’) structures—were, according to evolutionary theory, principally the causal by-product of environmental-*cum*-populational factors acting upon heritable developmental systems to produce adaptive modifications. While its ontological basis remained empirically and experimentally elusive during Darwin’s time, his emphasis on the all-pervading influence of heritable adaptation was vindicated with the rediscovery of Mendel’s ‘laws’ and, more importantly, the detailed detection of the molecular structure of DNA.¹¹ The genome, comprised of individually discernible, discretely modifiable trait-building units of inheritance, provided evolutionary theory with a robust conceptual framework with which to explain morphogenesis.

That framework—known today as the ‘Modern Synthesis’—was forged in the early 20th century and represented the operative integration of the genomic basis of heritable morphological modification with the populational propagation mechanisms of Darwinian natural selection within an overarching schema with which to explain the presence (and absence) of organismal features.¹² In the context of natural selection, properly understanding and explaining the process of evolution was taken to necessarily involve what Mayr (1975) influentially described as ‘population thinking’—that is, a methodological focus on *groups* of organisms (rather than *individual* organisms) as the principal participants in that process.¹³ And with the discovery of the developmental centrality of DNA, that process could now be understood in a novel fashion—namely, as changes in the genomic composition of organism populations over time (Dobzhansky 1937).

In the Modern Synthesis, explanations of the operation and effects of natural selection could then use a well-understood theoretical tool of physics and mathematics, namely, statistical analysis. By abstractly conceptualising organism populations as ‘gene pools’, the

morphological diversity produced by the process of evolution could now be quantitatively modelled as (and measured by) the distributional changes in gene frequencies within those pools over time.¹⁴ On the new explanatory framework of the Modern Synthesis then, the goal-directed morphogenesis of organismal features was no longer an intractable and inexplicable phenomena—it is, in the end, nothing more than the adaptive by-product of undirected populational variation and a mere reflection of selective pressures filtering the random rippling of gene pools.

That the genetic theory of natural selection central to the Modern Synthesis powerfully carried forward the Darwinian project of effectively purging the biological sciences of their need to appeal to teleological causes is well known. But what is less often considered, and what I want to illustrate here, is that this isn't as modest an excision as it's typically made out to be: for the explanatory structure of the Modern Synthesis doesn't simply reject one of the posits of Aristotelian ontology—it suggests a full-scale upending of its core metaphysical principles. For note that, according to the Modern Synthesis, the shaping of organismal morphology is a fundamentally *extra-organismal* affair. On that framework, properly explaining the presence or absence of an organism's features requires appealing to populations and the environmental stressors which act upon them, as it is variations in these factors which function as the significantly determinative *difference-makers* of morphological differentiation. Thus because whatever proximate contributions it might make to the construction of its own morphology are ultimately theoretically irrelevant (Mayr 1961), it is the extrinsic selective processes in which organism populations inadvertently partake that have genuine causal-*cum*-creative control over the specific features that make up any particular organism's morphological constitution.¹⁵

From the perspective of the Modern Synthesis then, because organisms are intrinsically inert participants in the formation of their own morphologies—it being a process in which they are passively 'pushed' and 'pulled' by the selective pressures within their respective populations—the very modality of morphology is a fundamentally *undirected* and *accidental* phenomena. For according to the framework of the Modern Synthesis, which features organisms *could* or *must* have is principally determined by the combination of intrinsic chance and extrinsic accidents. The 'drift' and 'flow' of the genetic variation which form the 'raw material' for the selective processes shaping organismal morphology have no privileged probabilities. Which pathways those processes follow in any particular instance is merely the result of a population's incidental navigation of whichever biotic or abiotic environmental pressures have happened to achieve a sufficient measure of proximal predominance.¹⁶ Thus, the apparent restrictions on the 'space of possibilities' exhibited in the stability of organismal morphology throughout the phylogenetic trajectory of a particular population of the sort traditionally conceptualised as

an effect of the normative constraints of kind-specific goal-directedness must be, according to the Modern Synthesis, a metaphysically derivative phenomena, and its instances simply ‘frozen accidents’, little more than artefactual silt shaped by the perpetual ebb and flow of selective processes.

In sum then, the utilization of the explanatory structure of the Modern Synthesis naturally suggests the adoption of a corresponding *metaphysical* perspective on the nature of causation according to which organisms and their morphological features are but the powerless playthings of evolutionary processes, *passively* and *extrinsically moulded* by *accidental* forces. This should come as no surprise: for because, as should by now be clear, properly accounting for an organism’s morphology—and the representative morphologies of natural kinds (if there are any)—requires conceptualising it as the product primarily of that organism’s place within the wider context in which it happens to be embedded and the relations it bears to other entities therein, the Modern Synthesis implicitly operates on a broadly *mechanistic* concept of explanation.¹⁷ It is this intimate interdependence of *explanation* and *ontology* that I hope the preceding discussion has highlighted: with the Modern Synthesis’s adoption of a mechanistic explanatory framework comes its denial of an Aristotelian metaphysics, and thus its rejection of the efficacy and existence of *formal causation*.

3 A More Modern Synthesis

It’s no exaggeration to say that the explanatory structure of the Modern Synthesis has undergirded one of the most successful research programmes in the history of the biological sciences and, given the intimate relation between explanation and ontology, it’s therefore no wonder that Aristotelian metaphysics have been so virulently rejected in its wake: if our evolutionary explanations of organismal morphology are manifestly bereft of teleology, so too should be our ontology of organisms. However, while the conceptual link between the acceptance of evolutionary theory and the dismissal of formal causation has been understandably persuasive, it is not unassailable: for the strength of that connection depends upon its being made in the context of the Modern Synthesis, and the Modern Synthesis is not only no longer the most modern synthesis, but its explanatory framework is no longer the only operative underpinning of evolutionary theory.

Indeed, since the latter half of the 20th century the omnipotence of the extrinsic and accidental mechanisms of natural selection to account for organismal morphology enshrined in the ‘adapationist’ paradigm of the Modern Synthesis has been continually thrown into serious question. For it has become increasingly apparent that its explanatory framework is unable to adequately account for a wide range of evolutionarily

important morphological phenomena – including (but not limited to) the discernible biases in character variation within phylogenetic trajectories, and the origin, prevalence, and persistence of homological structures in both parallel and convergent phylogenies. It is now rather evident that this inability is no mere theoretical idiosyncrasy, but rather stems from an essential oversight within the Modern Synthesis perspective: its systematic neglect of the generative architecture responsible for organismal ontogenesis (Müller and Newman 2003; Laubichler and Maienschein 2007; Huneman and Walsh 2017).¹⁸ In the new explanatory framework of evolutionary developmental biology (evo-devo), the selective forces of an organism's environment and their adaptive influence upon the allele dynamics of its populational gene pool are relegated from their reign and the generative capacities of intra-organismic developmental systems positioned in a place of primacy.¹⁹

In essence, by emphasizing the role that these developmental systems play within the process of evolution, the explanatory structure of evo-devo effectively collapses the proximate/ultimate distinction so central to the Modern Synthesis's account of the formation and transformation of organismal morphology.²⁰ And that role is, according to evo-devo, a significant one: for it is the dynamical features of these systems which are understood as shaping the possible 'playing field' of and directing the probabilistic trajectories of the process of natural selection. In the explanatory framework of evo-devo, this two-fold role is modelled by the interplay between evolvability and developmental constraint.²¹

With respect to the former, evo-devo highlights the prowess of the ontogenetic architecture responsible for the modular development of individual organismal traits in grounding the generation of heritable morphological variation (Hendrikse et al. 2007). As empirical research has revealed, the generative competency of the intra-organismic systems responsible for 'trait-building' is not limited to the production of a *single* and *specific* form: as expressed in the measures of their phenotypic plasticity—environmentally correlated quantitative and qualitative alterations in form (Pigliucci 2001; Fusco and Minelli 2010)—the dynamical features of these systems are capable of generating a multifaceted repertoire of permutations on their associated trait—an entire 'morphospace'.²² From an evo-devo perspective, what is important about this empirical-*cum*-theoretical insight is that it is the generative potential for phenotypic variation encompassed in the morphospaces of developmental systems which provide the 'raw material' upon which the vagaries of the environment and the process of natural selection may subsequently act. In other words, because adaptation can only operate on what is available, and what is available is a causal consequence of the dynamical features of those systems, the ontogenetic architecture of organisms plays a central role in defining the 'playing field' of the evolutionary process, and hence is of significant explanatory import in accounting

for the formation and transformation of organismal morphology (West-Eberhard 2003; Pfennig et al. 2010; Moczek et al. 2011).

While evo-devo emphasises the prowess developmental systems possess for generating adaptive phenotypic variation, it also highlights those systems' role in *constraining* that variation: for as empirical research has revealed, not all of the available permutations in any particular morphospace are equally available. This is because, in short, the dynamical features of an ontogenetic system from which such permutations arise also play a role in probabilistically biasing the generation of certain variations, rendering some less and others more likely to be regularly produced or maintained in the face of perturbation (Maynard Smith et al. 1985; Arthur 2004; Brakefield 2006).²³ From an evo-devo perspective, what is important about the biased structure of morphospaces is the role that the developmental constraints it imposes upon ontogenesis play in the process of natural selection. In particular, these constraints result in both *resistance to* and *restrictions on* morphological variability. On the one hand, because the dynamical biasing of morphospace can manifest in those systems' resistance to quantitative or qualitative alteration along certain dimensions of phenotypic space—as exhibited in the highly conserved morphological features in phylogenetic histories, e.g. homologues—these constraints 'drive' or 'channel' the selective process to proceed in particular ways (Arthur 2001; Brigandt 2015). On the other hand, as a consequence of this, the biased structure of morphospace can result in certain areas of that space being effectively 'forbidden' in such a way that some potential phenotypic variations which might be produced by those systems are rendered inaccessible by their developmental dynamics—a restrictive constraint which results in those areas remaining largely 'unexplored' by existing phylogenetic trajectories (Olson 2012).

There are, of course, many interesting theoretical implications of the explanatory structure of evo-devo, but for the purposes of this chapter I want to draw out what I consider to be the overarching theme which emerges from them: from the perspective of evo-devo, the shaping of organismal morphology in the evolutionary process isn't an entirely extrinsic and accidental affair in which organisms play a merely passive role. Indeed, as Müller (2007: 947) concisely puts it, 'evo-devo moves the focus of evolutionary explanation from the *external* and *contingent* to the *internal* and *inherent*.' Accordingly, when compared to the explanatory structure of the Modern Synthesis, evo-devo offers a radically distinct understanding of what I earlier termed the modality of morphology. For it posits that the central participants in the process of evolution are intra-organismic systems whose intrinsic dynamical structure both delivers and delimits its possible directions and thereby actively undergird the regularities which non-accidentally pattern its actual paths. Importantly then, in contrast to the Modern Synthesis, because the explanation of organismal morphology doesn't require conceptualising it as the product

of that organism's place within the wide context in which it happens to be embedded and the relation it bears to other entities therein, the explanatory structure of evo-devo does not construe organisms as merely 'cogs caught in the wheels of the evolutionary machine': the representative morphologies of natural kinds are not merely the resultants of the 'push and pull' of happenstance circumstances which befall their breeding pools, and the shape and structure of their phylogenetic lineages are not aleatory artefacts, but reflections of the generative dynamics of developmental systems' causal underwriting of a directed exploration of morphological space over evolutionary time-scales.

With this in mind, and recalling the conceptual move from explanation to ontology that the Modern Synthesis perspective makes in its rejection of Aristotelianism, we are now in a position to ask: given that the explanatory structure of evo-devo declares organismal morphology to be the *non-accidental* result of the *intrinsically dynamic* features of organisms, might it thereby offer support for the existence of non-mechanistic, *formal* causation? Of course if—even in the case of intra-organismal ontogenetic systems—it's 'mechanistic explanation all the way down', such support will not be forthcoming. In the next section, however, I argue that this isn't the case: for as I will illustrate, the evolutionary explanations of organismal morphology which are the centrepiece of the evo-devo framework principally appeal to the irreducibly non-compositional dynamics of goal-directed activity.

4 The Form of Evolutionary Explanations

I've argued that the key features to which the explanatory structure of evo-devo appeals in order to account for the origin and transformation of organismal morphology are two—evolability and constraint. As we have seen, the former concerns the generative *availability* of phenotypic variation (upon which selection can act), the latter the generative *accessibility* of that variation. To put it another way, the former concerns the 'reach' of organismal morphology, defining which variants are *possible*, while the latter concerns the 'restriction' of organismal morphology, delimiting which variants are *privileged*. Taken together these two features account for the 'directedness' of organismal morphology over evolutionary time-scales, as it is their interplay which performs the fundamental role of underpinning the various convergences and divergences of phylogenetic pathways. The pertinent question is then whether these two features are explicable via mechanisms—that is, whether these features can be accounted for by appealing to the *mechanistic* causal structure of intra-organismal ontogenetic systems.

Evaluating that claim requires getting clearer about the core features of the programme of mechanistic explanation. With the so-called new mechanism movement currently *en vogue* in the philosophy of science, there are

quite a few fairly nuanced definitions of mechanisms in the ever-growing literature on the topic, and thus many accounts of what mechanistic explanation amounts to (see Glennan and Illari 2018). For our purposes, we can take a mechanism to be ‘a structure performing a function in virtue of its component parts, component operations, and their organisation.’ (Bechtel and Abrahamsen 2005: 423) Mechanistic explanation thus has two steps. The first, and most fundamental, is the twinned process of decomposition and localisation whereby systems are ‘broken down’ into collections of spatio-temporally discrete elements (decomposition) and each individually assigned unique functional roles (localisation).²⁴ Once such systems are dissected into mereological sums of entities and activities (Machamer et al. 2000), the second step consists in temporally tracing causal trajectories through their structural interconnections—via their localised functional roles—to an end-point product (i.e. the *explanandum* phenomenon). In alignment with the previous historical discussion, we can say that mechanistic explanation involves conceptualising the phenomena to be explained as the functional product of a collection of entities being embedded within a spatio-temporally ordered web of extrinsic relations in which they are causally connected.

Although mechanistic models are prevalent in nearly every area of the biological sciences, what is important for present purposes is the fact that they feature prominently in explanations of the development and evolution of organismal morphology. The ontogenetic systems causally responsible for that morphology are typically taken to be genetic regulatory networks (GRNs) which consist of hierarchically ordered sets of genes, their transcriptional products, and their cross-regulatory interactions. Very roughly, genetic regulatory networks are composed of multiple sub-circuits whose products exert spatial and temporal regulatory control (via the activities of transcription factors) over the expression of various ‘downstream’ target gene batteries whose products then directly specify cell-type fate and are thus responsible for the three-dimensional construction of particular morphological features. The expressional interdependence among the genes in a genetic regulatory network whose regulative interplay results in the ontogenesis of such features is quite naturally modelled mechanistically as a spatio-temporally stratified edge-structure in a directed Boolean graph (Davidson and Peter 2015); see Figure 12.1a. In the context of development then, mechanistic explanation via genetic regulatory networks involves invoking the ‘regulatory logic’ of these networks in the traversal of specific sets of signalling pathways along a successive (though often non-linear) series of edges.

The practice of modelling the genetic regulatory networks responsible for the development of organismal morphology mechanistically has been incredibly successful and of great importance to experimental research in evo-devo (Wagner et al. 2000; Hall 2003; Laubichler 2010).

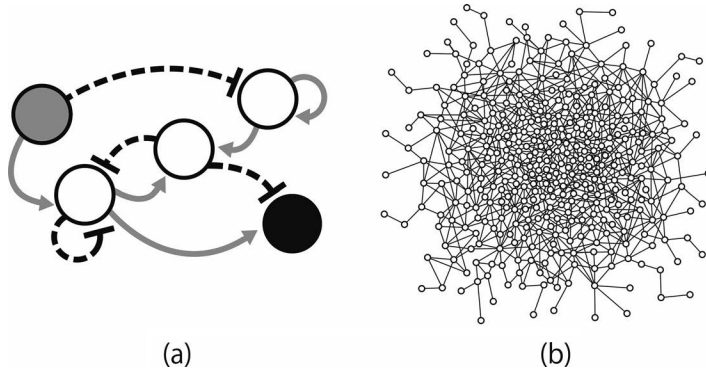


Figure 12.1 The structure of genetic regulatory networks. (a) Simplified structure of a genetic regulatory network depicting individual genes as discrete elements (or nodes) connected by various sorts of arrows (or edges) which each represent distinct sorts of transcriptional influence (on/off-higher/lower, etc.); (b) complex, computation-derived genetic regulatory network structure represented in a so-called 'hairball graph'.

However, the problem with these models and their associated explanatory insight is not just that their overwhelming complexity often fails to make the causal structure of the process of development perspicuous (see Figure 12.1b), but that, as many theoretical biologists are now appreciating, their exhaustively compartmentalised schematic is fairly inept at representing *system-wide* features of that process—in these models, as Huang (2011: 2247) concisely puts it, '[e]ntirety of analysis has trumped analysis of entirety.'²⁵ This is important because the aforementioned aspects of the process of development to which evo-devo makes explanatory appeal are such features: they represent the 'holistic' and integrative dynamics of those systems.

In the context of developmental biology, the most successful methods of modelling these features utilize the conceptual tools of dynamical systems theory (DST).²⁶ Rather than modelling the process of development in a modular ontogenetic system as a step-wise causal progression through a series of discrete state-changes in a collection of isolatable elements, dynamical systems theory takes a more comprehensive approach: its models represent that process as a continuous series of temporally successive transitions between entire system states. More specifically, the process of development in dynamical systems theory is modelled by tracing spatio-temporal trajectories through an abstract, multi-dimensional map—a state space—whose coordinate points represent possible whole system states, each of which reflect a complete 'genetic expression profile' of the system. Because the transitions between the states within that

space are governed by the system's 'regulatory logic', each state can be assigned a *stability value*—a contextual measure of the likelihood that the system in that state will shift to one of its neighbouring states. When these values are represented by an additional 'elevation' dimension (where a higher elevation denotes a greater degree of regulatory instability), our abstract state space becomes a structured topology, complete with peaked hills and low-lying valleys (Wang et al. 2011; Huang 2012; Davila-Velderrain et al. 2015; see Figure 12.2).

With the aid of dynamical systems theory, we can model the process of the development of a particular morphological feature in a novel fashion: the state of the ontogenetic system at any time can be treated as a kind of frictionless orb and the temporal succession of various distinct states of the system throughout that process conceptualised as the dynamic trajectory that orb traverses along a pathway geometrically guided by the topological curvatures of the system's state space—from the heights of the system-wide instability of its initial conditions 'downward' toward a state of regulatory stability; as represented in Figure 12.2. What is important for present purposes is that, unlike models based on genetic regulatory networks, the models of dynamical systems theory are capable of clearly representing the two key developmental features appealed to in evo-devo explanations: evolvability and constraint.

With respect to the former, by utilizing the data derived from experimental evidence of the states of systems which contribute to or themselves constitute system-wide regulatory stability, the models of

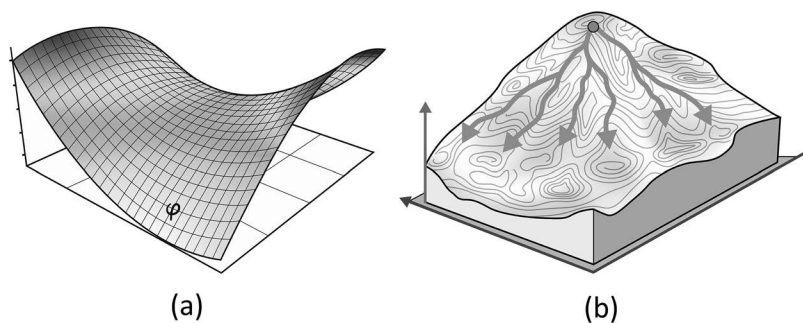


Figure 12.2 Topological structure of state space. (a) partial topological model of state space representing system-wide regulatory stability as a measure of elevation value, the ϕ -region signifying a set of quantitatively similar 'expression profiles'; (b) more comprehensive topological model of state space wherein the orb represents the system's initial, undifferentiated expression profile, the arrows distinct developmental trajectories through state space, and the low-lying contours at the arrows' termination stable attractor basins, each of which correspond to a quantitatively and qualitatively similar set of expression profiles.

dynamical systems theory represent a system's comprehensive capacity for phenotypic variation, depicting the developmental availability of the morphological permutations which comprise its morphospace as low-lying basins of attraction in the topology of that system's state space (Enver et al. 2009; Huang and Kauffman 2012). With respect to the latter, the models of dynamical systems theory also represent the causal-*cum*-structural 'shape' of that capacity by depicting the developmental privileging of and constraint on those permutations as the consequence of the topological curvatures of that system's state space (Huang 2009). There are two dual phenomena of developmental directedness: pleonasy, that is, the system's bringing about an end-state via a number of alternative trajectories, and persistence, the system's maintenance of an end-state through alterations of its trajectory. Both of these are represented in these models:²⁷ the first by the *width* of the various valleys in that topology, the second by the *steep gradients* surrounding their basins; see Figure 12.3. Taken together, these phenomena illustrate that system's resistance to and restriction on morphological variation.

Modelling the causal structure of an ontogenetic system with the conceptual tools of dynamical systems theory thus allows us to capture both the *plasticity* and *robustness* of their generative competency—the two features which, as I earlier claimed, distinguish the explanatory structure of evo-devo from the Modern Synthesis with respect to the modality of morphology. We are now in a position to evaluate the question with which this section began—namely, whether these features are mechanistically explicable. According to the schematic of dynamical systems theory which is required to represent these features, it is plausible that they are *not*: for they are neither the product of, nor amenable to the process of decomposition and localisation.²⁸

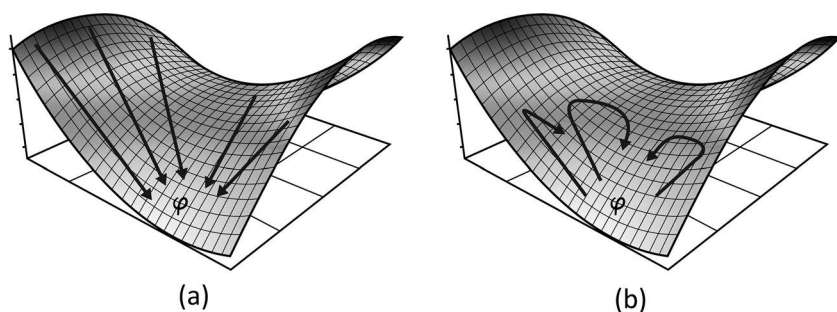


Figure 12.3 Topological features. (a) Pleonasy (also known as equifinality)—the width of the attractor basin in state space (ϕ) ensures its being 'reachable' by multiple distinct developmental trajectories; (b) Persistence—the curvature and depth of the attractor basin ensures that trajectories which result from regulatory perturbations regularly fail to 'escape' the region.

To see that this is so, note firstly that the topological structure in which both features are grounded is itself not a compositional element of an ontogenetic system, or strictly reducible to any such element (or set thereof). Given that its coordinates consist in a set of comprehensive whole-system states which form a continuous and exhaustive mapping of every available iteration of the values of those states, it is arguably not a proper part of such systems or capable of being discretely anatomised into modular elements which could be bijectively assigned to the entities and activities of such systems; in these models, the *possible* outstrips the *actual*. Clearly then, the topological structures which model the plasticity of these systems are not derived from a typical process of decomposition—indeed, quite the opposite is the case: they are formed from the integrated consolidation of collections of cumulative system states.

With this in mind, note also that it is the geometrical features of these structures—and in particular, the metric-bending basins of attraction whose contours shape the sloped curvature of state space—which represent the relevant dynamics of a system with respect to its privileging and prohibiting of particular developmental trajectories. Given that the topology of a system's state space is the product of an integrative series of vector-summed stability measures in such a way that each point within that space is intimately connected to, and 'weighted' with respect to, its neighbouring points, it's clear that the gradients which generate the geometry of that space cannot be attributed to any particular spatio-temporal configuration of the system's constituents and their activities. Thus, representing the phenomena of developmental constraint requires invoking the holistic dynamical 'flow' of the topology of a system's state space—the course of which cannot be derived from a typical process of localisation, nor be itself properly localised.

In short then, the *plasticity* and *robustness* of the developmental processes in which the ontogenetic architecture of organisms engage are grounded in essentially *system-level* features. For this reason they are in an important sense mechanistically inexplicable—indeed, the explanatory efficacy of the models which depict them rests upon their representing the accumulative integration of the entire 'dynamical repertoire' of those systems. And importantly, it is precisely this aspect of such models to which the explanatory structure of evo-devo makes principal appeal—for in their rendering of the structure of state space, they provide 'a map of the possible (and probable): a substrate on which evolution by natural selection can work.' (Jaeger and Sharpe (2014: 73); compare with Stadler et al. (2001)) With these points in mind what I want to now suggest is that, in as much as these non-mechanistically explicable morphospace mappings reflect the intra-organismal, holistic and goal-directed developmental dynamics which ground the 'modality of morphology', these models represent *formal* features of ontogenetic systems.

As explicated in the first section of this chapter, in the Aristotelian framework the concept of ‘what it is to be’ an entity is not inert, or lifeless—with respect to form, the *definitional* and *dynamical* are one.²⁹ Indeed, according to that framework the essence of an entity is ineliminably teleological: it not only defines the ‘limit and *logos*’ of its morphology (*DA* II.4, 416a15–9) but directs its ‘process of development ... [to be] necessarily such as it is.’ (*PA* I.1, 640b1–4; both translations from Aristotle 1984) It is this aspect of form which, as we saw earlier, is meant to ensure the successes of the invocation of the form of an entity to answer ‘why?’ questions concerning that entity’s possession of particular features, or properties. We can now note that the structure of such explanations is remarkably similar to the one underpinning the explanations characteristic of evo-devo: each invoke the generative capacity of the dynamical architecture of organisms to establish the scope of their possible developmental trajectories and the extent to which each are probabilistically privileged.

If then, as I have illustrated in this section, the dynamics of this modal-*cum*-teleological function is represented in the structural features of morphospace, there’s every reason to think that such models depict—at least to some approximation—the *formal* features of organisms.³⁰ For not only do these higher-order, irreducibly goal-directed dynamics play an important developmental role with respect to morphogenesis, but they also provide a plausible basis for attributions of *kind-identity*: as represented generally in the recent resurgence of process ontology in the philosophy of biology (Henning and Scarfe 2013; Nicholson and Dupré 2018) and more specifically in the various attempts within contemporary theoretical and experimental biology to delineate cell-types (Enver et al. 2009; Bhattacharya et al. 2011; Li and Wang 2013) and homologues (Striedter 1998; Hallgrímsson et al. 2012; Jaeger and Monk 2014) by their ‘phase portraits’, the definitional and the dynamical are increasingly seen as being intimately intertwined.³¹ According to the framework I’ve elaborated throughout this chapter then, as the ontological ground of explanations of an entity’s property possession which cite the role its essence plays in bringing those properties about, formal causation is no philosophical phantasm: its prowess can be both coherently illustrated and shown to play an important theoretical role within evolutionary biology.

5 A Coda: Causation and Explanation

Thus far I’ve argued that explaining the key evolutionary phenomena of organismal morphology in the framework of evo-devo involves (a) conceptualising organisms as centres of *intrinsic activity* and (b) identifying their kind-defining attributes with their *system-level dynamical structure*. This is because, taken together, (a) and (b) give ground

to the features which are principally responsible for the modality of morphology—evolvability and developmental constraint. To accept (a) is to reject a conception of ontogenesis as primarily a *contingent* and *context-dependent* process in which organisms are themselves merely *passive* participants, and accepting (b) requires recognising the *non-decomposable* and *non-localisable* nature of the capacities which undergird that process. Accordingly, the explanatory framework of evo-devo is in an important respect indicative of the existence and operation of non-mechanistically explicable ‘formal causes’.

However, even accepting the arguments of this chapter thus far, one might still question whether this *explanatory* role really is either tantamount to, or else compelling evidence for the form of an organism *causing* it to possess certain properties. Granted—one might say—if we define the form of an organism in this fashion, we can invoke an organism’s kind-definition to *explain* its possession of developmental dispositions and thus *explain* some important evolutionary facts about the groups of organisms which belong to this kind—but where’s the *causation*? With the discussion of the previous section in mind, we can make this question more precise: what is *causal* about the ‘formal structure’ of morphospace? Here we are asking whether the geometrical gradients which comprise the topological structure of state space represent genuinely causal structure—do its steep slopes really *causally direct* and its wide basins really *causally constrain* the process of ontogenesis? The answer to this question, of course, depends upon one’s perspective on the metaphysics of causation, an independently and infamously complex issue.

From a broadly mechanistic point of view this question should be answered negatively. In that framework, as I earlier noted, causation consists in the spatio-temporal trajectories connecting organised collections of entities and activities. Given that, for reasons discussed in the previous section, the dynamical structure represented in morphospace models is not amenable to analyses via entities and activities it is easy to see why, from the mechanistic perspective, such structure must be regarded as being fundamentally deficient: without compositionally discrete entities, it is ostensibly bereft of any proper causal *relata*, and without functionally localisable activities, it fails to feature any discernible causal *relations*. Models which suffer from these deficiencies are typically regarded by proponents of the mechanistic paradigm as being essentially ‘phenomenal’—that is, as providing purely (and merely) *descriptive* representations of system processes: even if such models possess explanatory and predictive prowess with respect to those processes, they, nevertheless, fundamentally fail to delineate the ontic *causal structure* of the systems which undergird them (Glennan 2005; Kaplan and Craver 2011; Craver 2014).

The relegation of morphospace models as ‘merely phenomenal’ would certainly be well warranted if it were the case that causal structure *just*

is mechanical structure—that is, if what it is to be a *causal* structure is to be an *entities and activities* structure. But this can't be right—for because, as Kistler (2009: 599–600) points out, 'the concept of causation is among the irreducible conceptual instruments of mechanistic analyses', the two are *not* identical: in fact, any explication of the latter fundamentally presupposes the utilization of the former (and not vice-versa). In what sense then might the formal structure of morphospace models, from the perspective of the mechanistic paradigm, be considered *non-causal*? A plausible reading of the literature suggests that the claim that such structures merely 'track the symptoms' of causation is largely based on the charge that the models in which they feature invoke the wrong *sort* of causality—one whose instances, if they deserve the appellation of 'cause' at all, do so only derivatively.

Following Hall (2004), a contrast is often made between two fundamentally distinct concepts of causation—difference-making and production.³² Roughly, according to difference-making concepts of causation, *c* causes *e* just in case *c*'s presence (or absence) 'makes a difference' to the occurrence or state of *e*. Following Lewis's (1973) formulation, 'making a difference' is typically analysed via the concept of *counterfactual dependence* and in many cases via its direct descendant, Woodward's (2003) concept of *manipulability*. Although this familiar and widely adopted concept of causation is notoriously susceptible to a number of particular problems—cases of preemption, of overdetermination, and so on—a more foundational worry one might have with it is that it is both too general and too permissive to be informative: counterfactual dependence is everywhere, and it's far from clear that what all of its instances have in common is causality.

It is this perceived lack of informative content which is at the heart of the proposal of the production concept of causation: counterfactual dependence may be an observationally sufficient criterion of causality, but it fails to lay bare the 'objective physical causal connections' of the sort 'which Hume sought in vain' (Salmon 1994:297; compare Glenan 1996). The 'secret connexion' between cause and effect about which Hume professed Socratic ignorance consists, according to production accounts of causation, in the generative activity of spatio-temporally extended processes (Salmon 1984; Dowe 2000). On this perspective, the relations tracked by counterfactual-based dependence analyses fail to capture the *oomph* of causation and are little more than conceptual 'book-keeping': if *c* 'makes a difference' to *e*, it is *because* the former actively engages in a process which brings about the latter.

A broadly mechanistic perspective on the nature of explanation might be understood as being rather naturally aligned with a productive concept of causation in that the integrated operation of the entities and activities of mechanisms plausibly constitute the diachronic activity of a generative process (Williamson 2011; Campaner 2013). With this in

mind, the relegation of morphospace models as merely ‘phenomenal’ might be justified by the claim that the formal causation they invoke is nothing more than mere difference-making—and thus *causal* in some derivative and residual sense only. Along these lines one might grant that the topological structure of morphospace *makes a difference* to the process of ontogenesis, and that those differences might themselves *make a difference* to the process of evolution but deny that its geometrical gradients deliver the propagative prowess which constitutes the fundamental ‘web of causal influence’ (Strevens 2007).

If there exists a genuine dichotomy between these distinct sorts of causation *and* the two are hierarchically ordered in the aforementioned fashion, the link between formal *explanation* and formal *causation* may not be as straightforward nor as strong as the neo-Aristotelian supposes. However, I maintain that this line of argument is not as compelling as the proponents of a mechanistic perspective might hope—for when subject to sufficient scrutiny, it is not at all clear that these two conditions can be so easily satisfied. One the one hand, the very existence of that dichotomy can be called into question. Although advocates of the production concept of causation typically denounce the dependence concept as being both metaphysically mysterious and essentially uninformative, they are plausibly open to a philosophical *tu quoque*. For while the production concept of causation is supposed to get at the genuine ‘web of causal influence’, giving an account of what exactly is *in* this web is more easily said than done.

It’s not uncommon here to see proponents of the production concept shift the onus of explanation onto the shoulders of the empirical sciences by insisting, as Ney (2009: 759) does, that the web is constituted by ‘the objective features of the world [which are] revealed by a physical theory of causation’. However, without wishing to disparage the enterprise of the natural sciences, it is safe to say that as of now no such revelation is forthcoming, and thus that this is an explanatory burden which likely will not be able to be discharged. It is well known that causation is a concept which is effectively treated as a kind of theoretical primitive in the empirical sciences and that, in those rare cases where it is subject to further analysis, it is often done so in ways the philosopher (and the metaphysician, especially) would judge to be either overly idiosyncratic or woefully imprecise. For, more often than not, when a more ontological interpretation of causation is present in the scientific literature, it is formulated *ex hypothesi*, and in service of supporting a particular theoretical interpretation of the relevant data, rather than arrived at by a non-ad hoc, evidenced-based extrapolation of the metaphysical make-up of a ‘web of causal influence’. Furthermore, although the more philosophically rigorous framework of counterfactual dependence is almost wholly absent from that literature, it is far from clear that the result of any such extrapolation is likely to reveal something substantially distinct from

various terminologically translatable forms of the difference-making relation. Advocates of a production account of causation are often wont to employ more empirical-sounding, Anscombe-inspired (1993) gerundives to describe singular causal events—binding, unwinding, transcribing, and so on—but any claim to the effect that the empirical sciences have imbued such terms with sufficient conceptual content which would warrant regarding them as anything other than disguised difference-making relations should be treated with suspicion.

More theoretical explications of the constituents of that web have arguably not fared much better. What precisely is a process, a supposed central component of this web? The answer that it is ‘a physical four-dimensional entity that has some distinguishing feature which makes it apt to be a means by which causal influence can be propagated’ (Handfield et al. 2008: 156) is I think, at best, uninformative. What then does causal influence amount to? I’ve seen no better attempt at a definition than that it is something which occurs when ‘a change in a property of one [thing] dynamically produces a change in a property of another [thing]’ (Tabery 2004)—true enough I imagine, but, nevertheless, rather plainly well below the bar of being philosophically illuminating. Difference-making may be a rather bare-boned conception of causation, but it’s highly questionable whether the various enflishments on offer in the literature are any more substantial.

Even if we were to suppose that the production concept of causation could be given enough substance to license its being sufficiently distinguished from the difference-making one, all the work of demonstrating that instances of the latter are fundamentally but shadows of those of the former yet lies ahead of us. And although advocates of production accounts routinely assume that such a demonstration *must* be possible, it has thus far failed to be sufficiently carried out, and there’s every reason to assume that this is no accident. For if there is a hierarchy of dependence which exists between production and difference-making, that relation rather plausibly runs the other way.³³ It’s difficult to imagine, for instance, that any coherent account of what it is for one thing to produce another can be properly explicated *without* employing the conceptual framework of counterfactual dependence: what is it, fundamentally, in virtue of which a process is *productive* if not its ability to ‘make a difference’ (Psillos 2004; Froeyman 2012; Glynn 2013)? Indeed, the utilization of that framework appears altogether necessary to properly individuate the components, identify the organisation, and construct the boundaries of mechanisms themselves—the supposed carriers of causal influence (Woodward 2004; Menzies 2012; Casini 2016; Ioannidis and Psillos 2018).

The charge that dependence represents a weaker, or less robust sort of causation is then, as far as I’m concerned, unfounded: its competitor neither substantially builds upon nor can itself be built without it.

The upshot of this discussion is that insofar as the formal structure of morphospace essentially consists in the functional integration of system-wide counterfactual dependencies, it arguably has every right to the claim of being genuinely *causal* that the ‘entities and activities’-based structure of mechanisms does. Furthermore, what this section has I hope shown is that even if one remains reticent to grant it such rights, formal causation—as explicated in the neo-Aristotelian framework I’ve proposed—should at the very least be regarded as being *no more mysterious* than whatever ‘oomph’, or ‘influence’ is, and its activity *no more miraculous* than whatever it is that difference-making does.³⁴ Thus if formal causation can be justifiably deemed esoteric, so too must efficient causation be, and for just the same reasons.

Conclusion

In this chapter I have illustrated the ways in which the mechanization of the natural world in the aftermath of the Scientific Revolution was crystallised in the biological sciences with the formation of the Modern Synthesis—a conceptual framework for evolutionary theory in which the teleological prowess of formal causation is rendered both explanatorily impotent and ontologically otiose. I have argued, however, that the radical reframing of that theory implicit in the explanatory structure of evolutionary developmental biology suggests that the Aristotelian posit of a non-mechanistic ground of the modality of morphology need not be treated with suspicion. Form may yet prove a metaphysically and empirically powerful apparatus with which to comprehend the nature of organisms in the context of our post-Darwinian picture of the living world.

Notes

- 1 See principally *Phys. Phys.* III.3, 194b27–9 and *Metaph.* V.2, 1013a24–b28. The connection between form and kind is discussed throughout *Metaphysics* VII–VIII.
- 2 This is not to say that Aristotle doesn’t sometimes *treat* the essence of an entity as a kind of ‘static’ definition (e.g. in the *Posterior Analytics*, and even *Metaph.* VII) or *utilise* it as such in logical demonstrations (Ferejohn 2013)—but this is an abstraction, and not to be confused with the *ontology* of Aristotle, wherein form and activity are inseparably linked (Kosman 2013).
- 3 This is the conclusion of the brief but famous ‘syllable argument’ of *Metaph.* VII.17, 1041b11–33.
- 4 See *Phys.* II.7 for the categorisation of the types of causes by the criteria of movement.
- 5 Recent examples include Walsh (2006, 2015), Nuno de la Rosa (2010), Boulter (2012), Lennox (2017) and Peterson (2018). Connell (2001), Oderberg (2007), and Devitt (2008). For my own contribution to this Aristotelian resurgence, see principally Austin (2018), but also Austin (2017a, 2017b).

- 6 Descartes—who found power explanatorily superfluous—and Malebranche—who judged them inconceivable, and worse, theologically inappropriate—are probably the most well-known detractors from a powers-based ontology in this period; see Clarke (2000) for a concise overview. Their epistemological arguments against powers were, of course, famously reiterated and reinforced by Hume's (1748/1999) analysis; see Bell (1997) for a fairly in-depth tracing of that philosophical lineage.
- 7 In Aristotle's metaphysical framework, the concept of teleological causation is indeed tightly bound-up with the concept of the 'soul' and its operations (as evidenced in *De Anima*)—and in the Early Modern period (for the reasons mentioned in the text) this association was ripe for ridicule; Malebranche's (1992: 227) quippy warning for those who wish to adopt an Aristotelian perspective on causation is always worth repeating: 'You would have, in short, to make a man out of an armchair.'
- 8 This is not to say that mechanistic theories of organismal growth and activity weren't proposed or discussed during this period—Descartes himself wrote plenty on the subject (see Des Chene 2001)—only that they weren't prominent (Wolfe 2011).
- 9 This suitability was, of course, first documented by Aristotle himself, in his *History of Animals*.
- 10 Cf. Mayr (2004): 'When one reads the writings of one of the leading vitalists like Driesch one is forced to agree with him that many of the basic problems of biology simply cannot be solved by a philosophy as that of Descartes, in which the organism is simply considered a machine. [...] The logic of the critique of the vitalists was impeccable.'
- 11 Though a theoretical lineage can be traced (Waters 1994), the molecularisation of the Mendelian 'gene' concept initiated by the discoveries of Watson and Crick was not as smooth a transition as some histories make it out to be, and it remains a much-discussed mutation (Falk 2010).
- 12 See Mayr and Provine (1998) for an excellent collection of papers on the historical and conceptual issues in the formation of the Modern Synthesis.
- 13 I note in passing that Mayr understood population thinking not merely as a methodological principle, but as an ontological postulate—for a discussion of the metaphysical dimensions of the concept, see Sober (1980) and Ariew (2008).
- 14 For an excellent account of the history of the development of the statistical methodology of population genetics, see Provine (2001).
- 15 The characterisation of natural selection as a *creative* process is admittedly contentious—see Beatty (2016) for a historical discussion on the subject.
- 16 For a historical overview of the concept of 'drift' and the theoretical issues surrounding its role in evolution, see Plutynski (2007).
- 17 This is not the claim that evolution itself—*qua* the process of natural selection—operates mechanistically *per se*, only that the formalisation of that process in the Modern Synthesis effectively mechanizes its explanatory role in organismal morphology. For more in-depth discussions on the machine-like activity of natural selection, see Skipper and Millstein (2005) and Barros (2008).
- 18 As Amundson (2005) expertly evidences, this neglect was not merely happenstance, but reflected the core commitments of the Modern Synthesis: in that framework, the causal structure of development is effectively 'black boxed', as whatever happens betwixt gene and form has no explanatory role to play in the statistical analysis of gene frequencies within organism populations.

- 19 Although there is a stark ideological contrast between the two frameworks, most evo-devo researchers view the discipline as an extension of the Modern Synthesis, or else a novel, though non-exclusionary synthesis of that synthesis with the study of developmental mechanisms—see Müller (2007), Pigliucci and Müller (2010) and Hall (2012).
- 20 For a wider-scoped contemporary critique of the tenability of Mayr's distinction, see Laland et al. (2011).
- 21 Although I have distinguished them here for the purposes of illustration, it should be noted that these two concepts are theoretically interdependent and the phenomena they represent are in a way two sides of a single coin—see Brigandt (2007, 2015) for an in-depth discussion.
- 22 On the general concept of morphospace and its theoretical applications, see Rasskin-Gutman (2005) and Mitteroecker and Huttegger (2009).
- 23 Demonstrating the importance of developmental constraints in the evolutionary process was an effort largely pioneered by the work of Pere Alberch during the 1980s—see Rasskin-Gutman and De Renzi (2009).
- 24 Cf. Bechtel and Richardson (2010) and Levy (2014).
- 25 Cf. Huang and Wikswo (2006), Jaeger and Monk (2015) and Brigandt et al. (2018).
- 26 Contemporary forms of dynamical systems theory in developmental biology represent the theoretical union between Kauffman's (1969) Boolean configuration of genetic regulatory networks and Waddington's (1957) 'epigenetic landscape' models. Models of development based on dynamical systems theory are widely applied in analyses of sub-organismal cell fate—in everything from the morphogenesis of plants (Álvarez-Buylla et al. 2008) to carcinogenesis (Kaneko 2011)—and have interesting applications in the study of the evolvability of organism populations (Striedter 1998; Jaeger and Monk 2014).
- 27 These markers for directedness were first proposed by Sommerhof (1950), who referred to them as 'plasticity' and 'persistence' respectively; as the current paper discusses the unrelated topic of phenotypic plasticity, I have substituted the term here. For recent examinations of these features in directed systems, see Nagel (1977) and Walsh (2012).
- 28 Cf. Kuhlmann (2011) and Silberstein and Chemero (2013).
- 29 The conceptual interdependence between being and activity in Aristotelian metaphysics—deftly illustrated by Kosman (2013)—is why he might be seen as 'the first functionalist' (Shields 1990).
- 30 It's worth noting that others have occasionally glimpsed this sort of association. In their discussion of the framework of evo-devo, for instance, Von Dassow and Munro (1999: 310) briefly—but explicitly—highlight the conceptual similarity between the causal privileging of morphology illustrated in models of dynamical systems theory and the 'finalistic' dynamics which typify formal causation. Kauffman (2014) identifies formal causes as 'boundary conditions' and 'enabling constraints' in state space schemas, and elsewhere (manuscript: 8) refers to the 'opportunity for adaptation' as a 'blind final cause'.
- 31 The affinity between dynamical systems theory and process ontology runs deeper than one might think: C.H. Waddington—the progenitor of topological analyses of ontogenesis—was deeply influenced by A.N. Whitehead—the philosophical figurehead of process ontology in the Western analytic tradition—as he himself notes (Waddington 1975).
- 32 What follows is only a truncated outline of these competing concepts of causation and the discussion assumes a degree of familiarity with the debate. For a more comprehensive examination see Beebe et al. (2009).

- 33 I do not here delve into the open option of simply denying the existence of such a hierarchy full stop (Strevens 2013; Glennan 2017)—though opting for this sort of position is obviously no boon for those who wish to declare ‘dependence’ to be an only derivatively causal concept.
- 34 For another recent exploration of the utility of formal causation *qua* difference-making, see Sandstad and Jansen (2021, this volume).

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